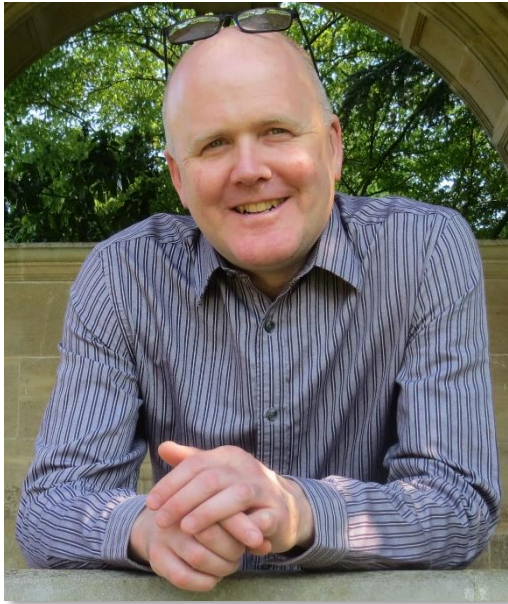
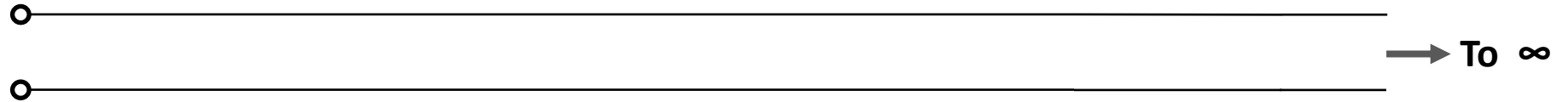


Reflections and Terminations

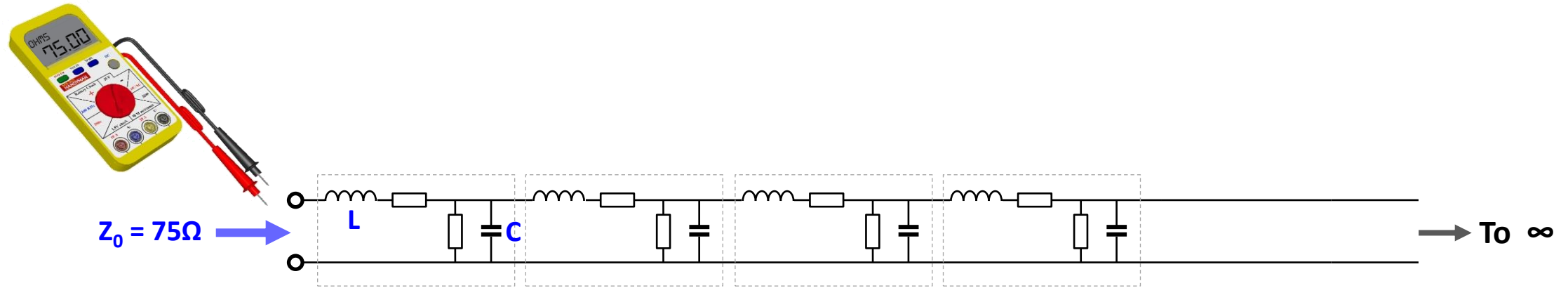


Mike Dhonau

Characteristic Impedance and Terminations



Characteristic Impedance and Terminations

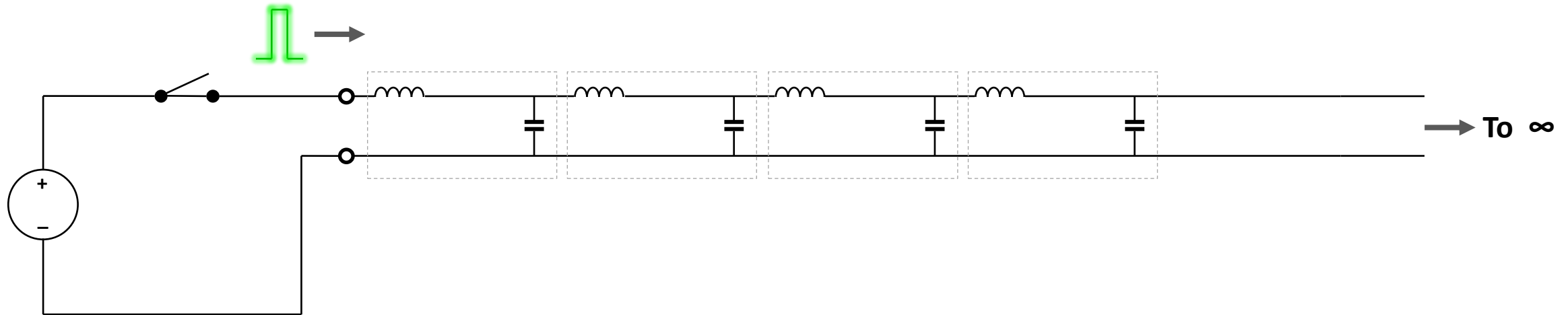


Characteristic Impedance:

$$Z_0 = \sqrt{\frac{L}{C}}$$

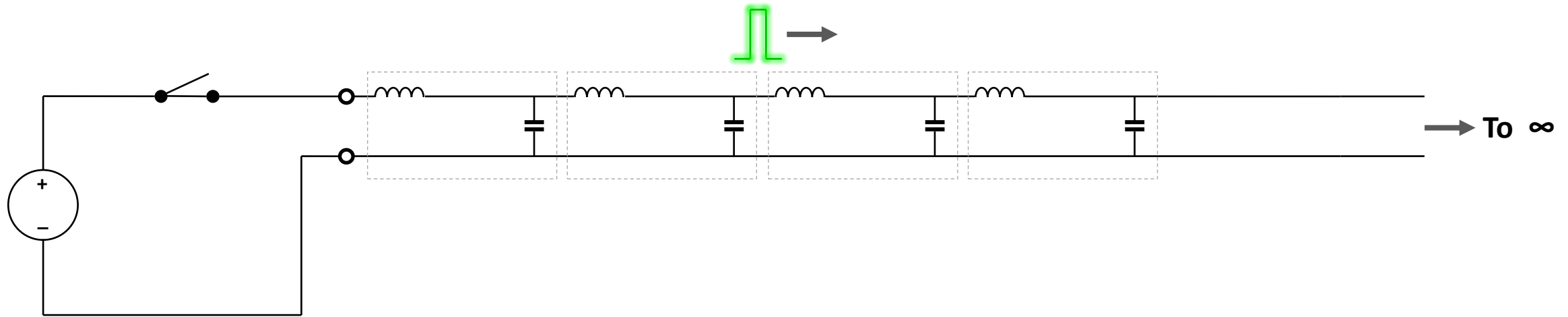
Characteristic Impedance and Terminations

1



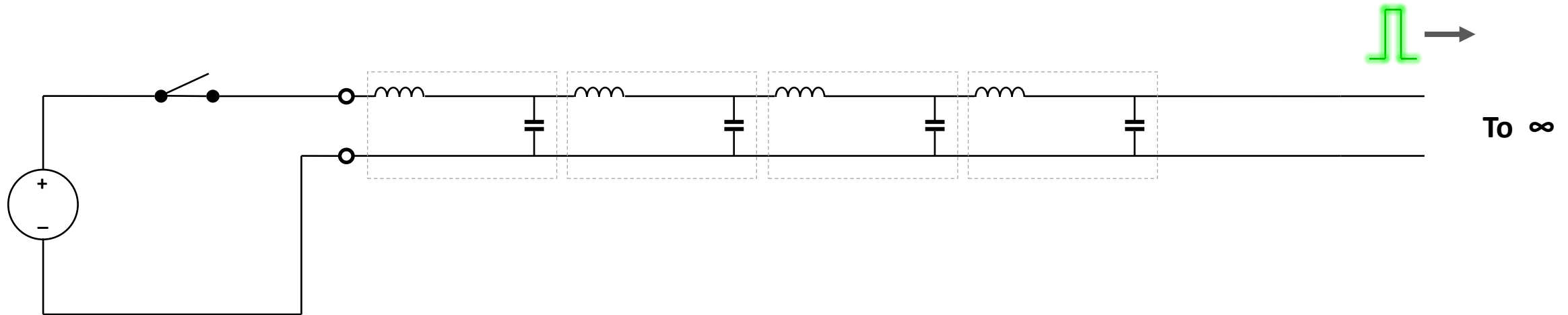
Characteristic Impedance and Terminations

2

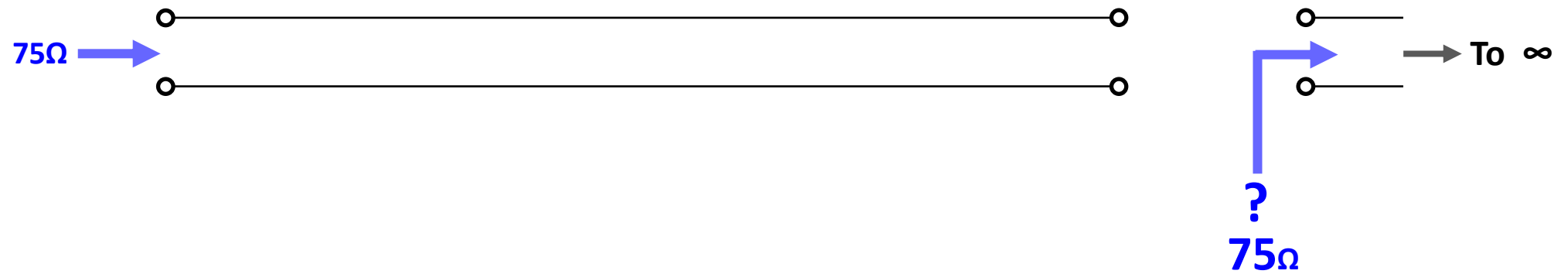


Characteristic Impedance and Terminations

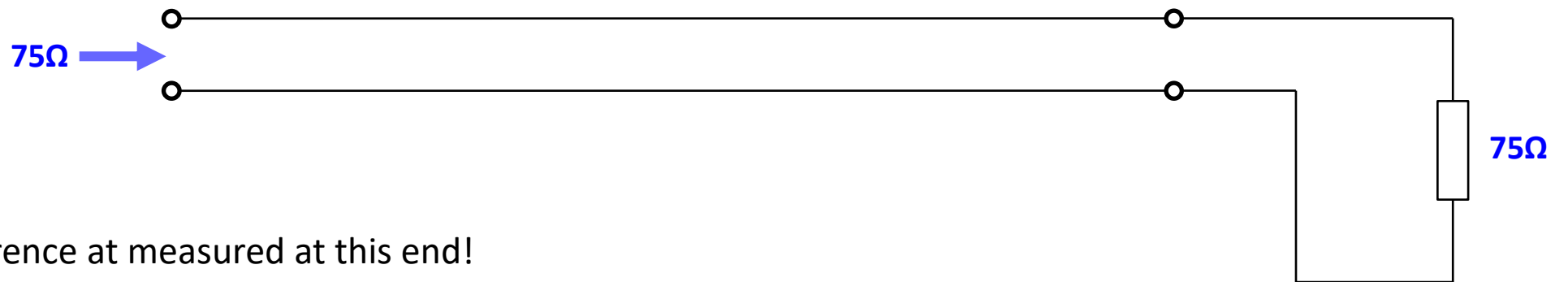
3



Characteristic Impedance and Terminations



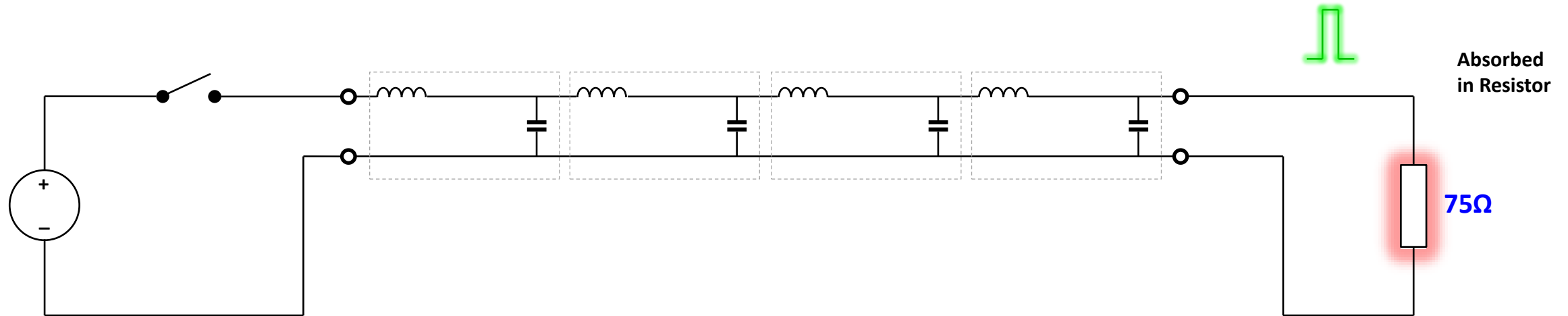
Characteristic Impedance and Terminations



No Difference at measured at this end!

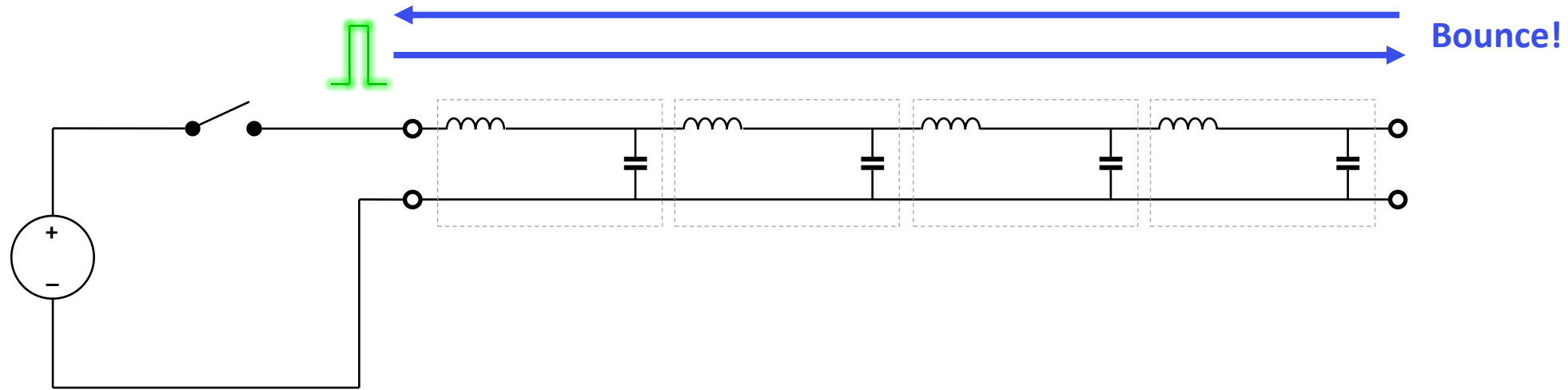
Therefore energy sent still goes to 'infinity'
(is actually absorbed in the termination).

Characteristic Impedance and Terminations



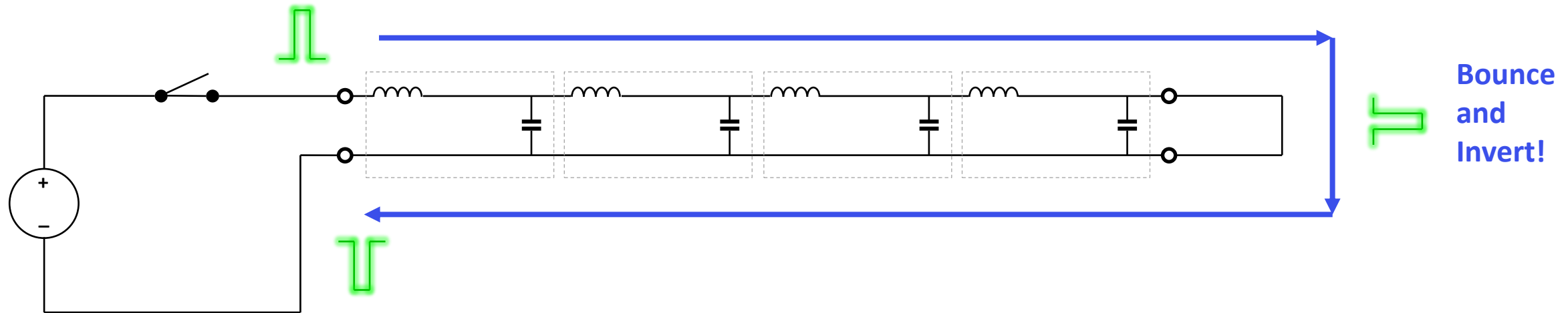
Correct Termination

Characteristic Impedance and Terminations



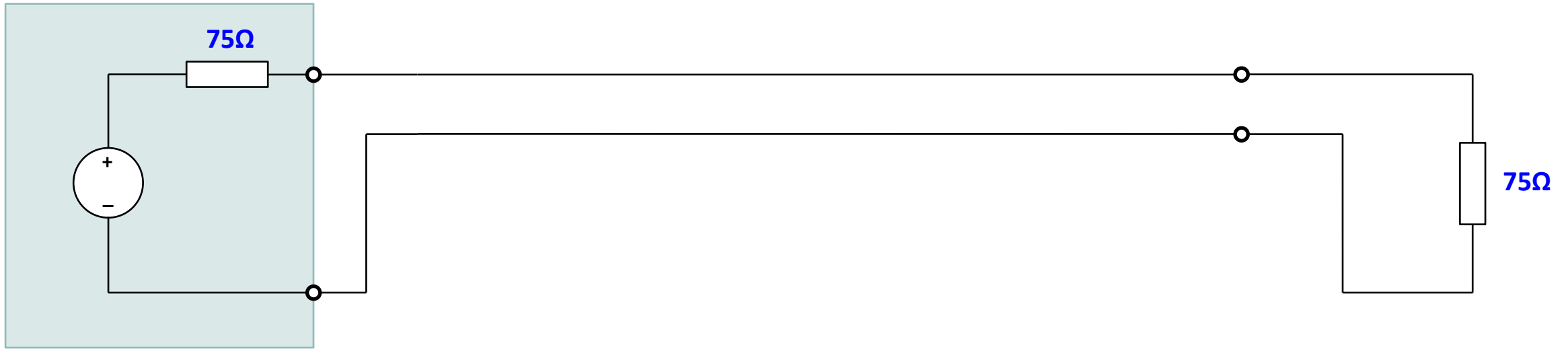
Open Circuit Termination

Characteristic Impedance and Terminations

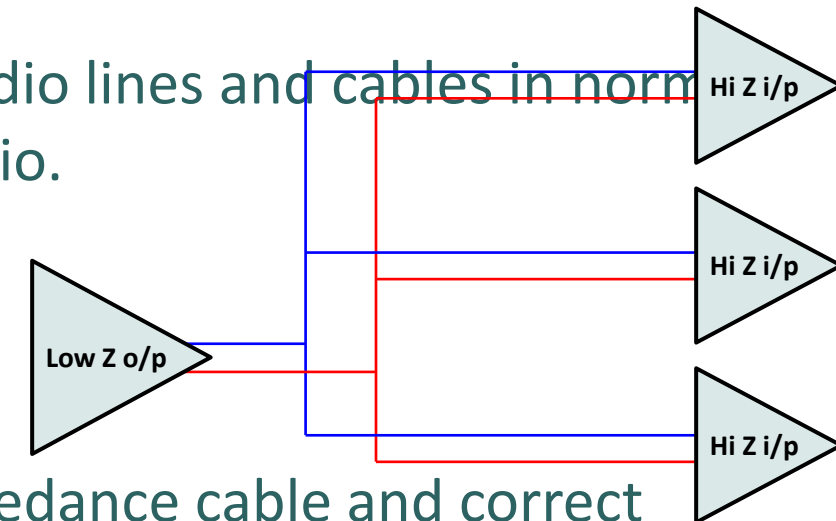


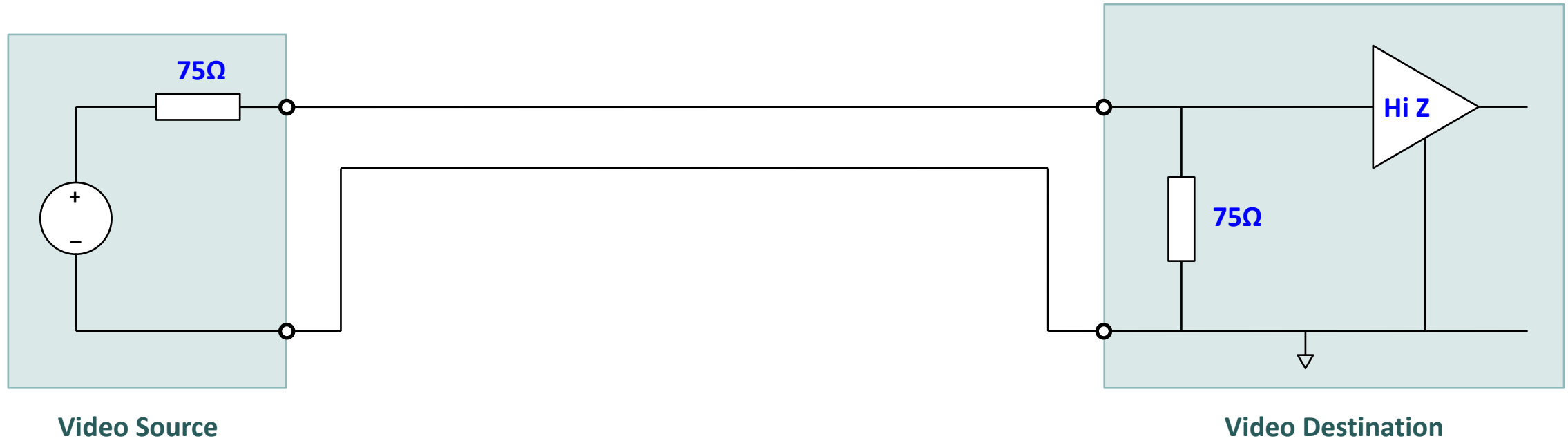
Short Circuit Termination

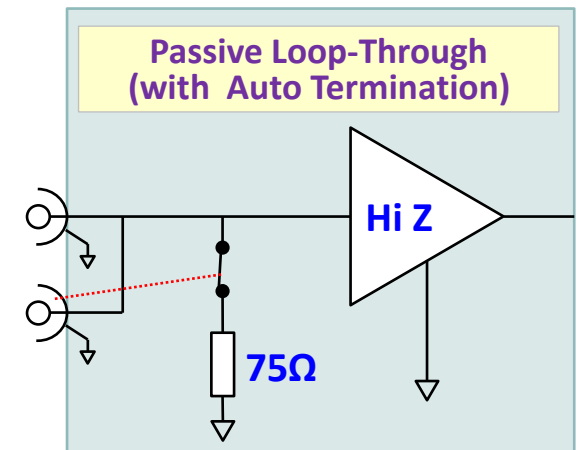
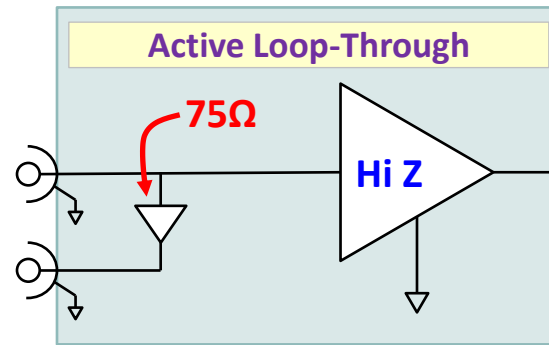
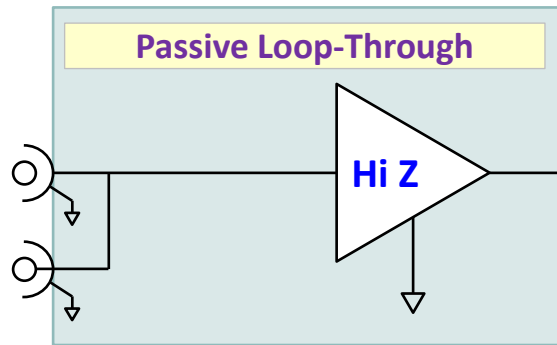
Correct Termination at Both Ends

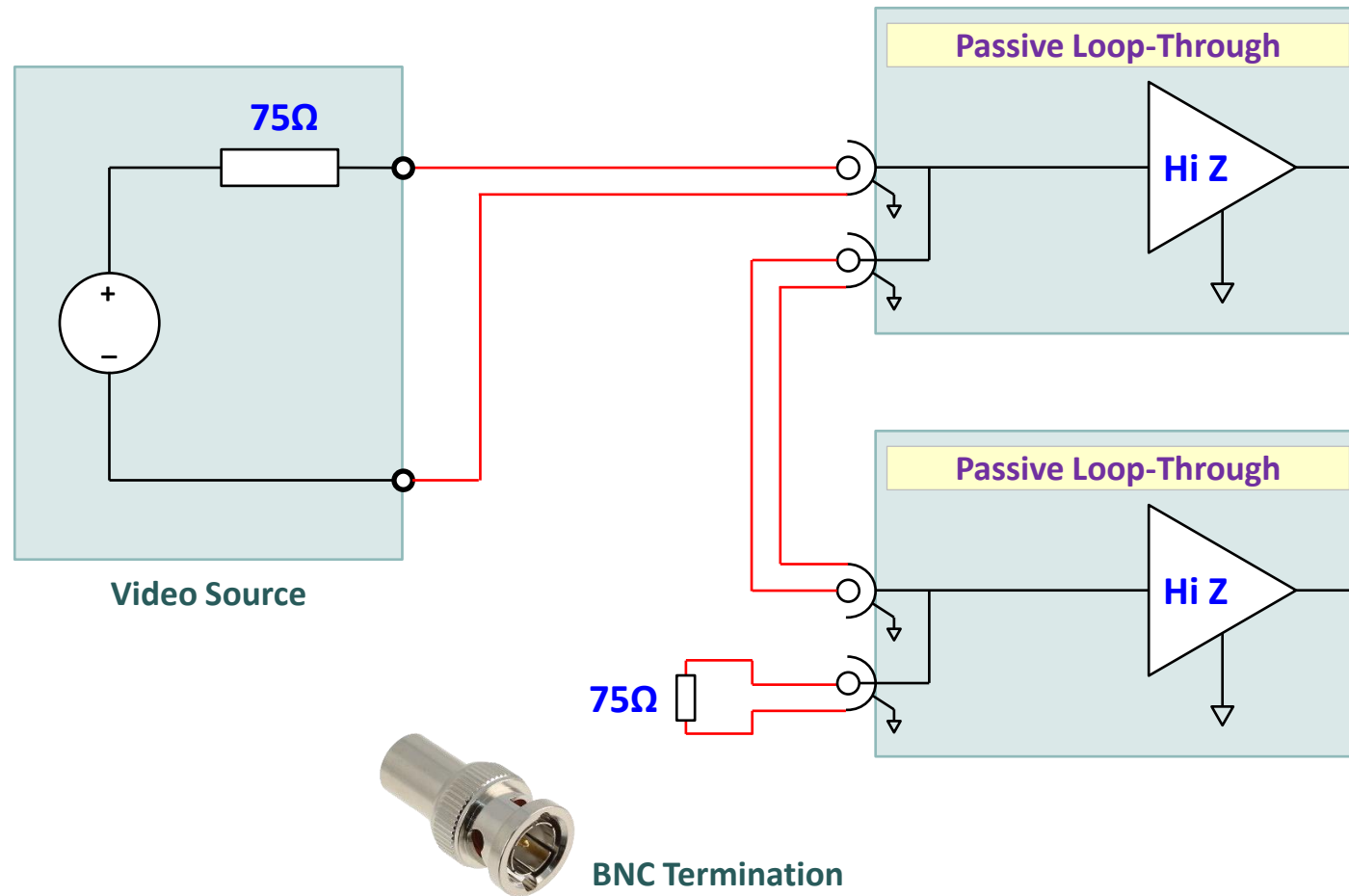


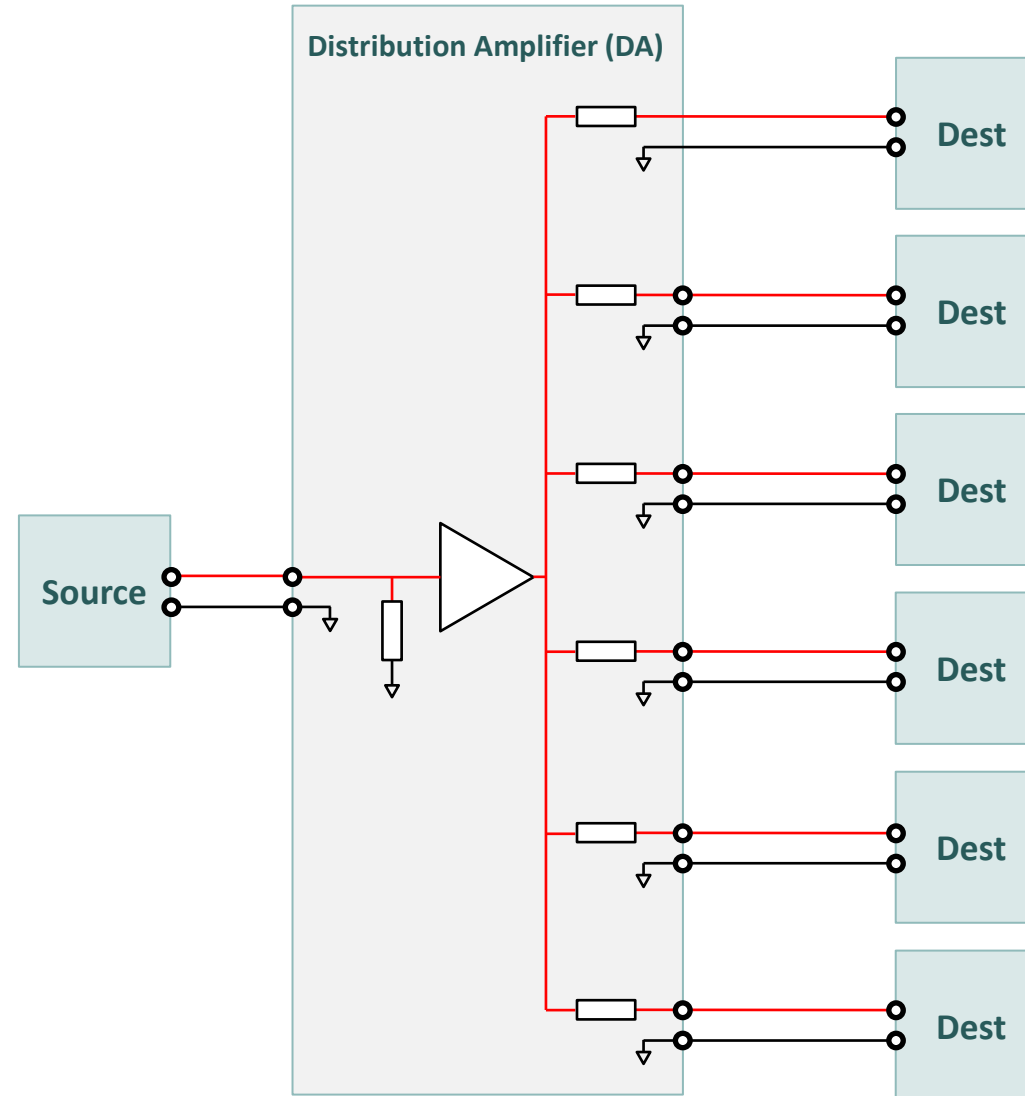
- All **professional audio** equipment has **balanced** inputs and outputs.
- Most equipment has the following impedances.
 - **High input impedance** (typically $> 10\text{ k}\Omega$)
 - **Low output impedance** (typically $< 600\ \Omega$)
- **Reflections do not matter** because the **round trip time** for audio lines and cables in normal use is very **short** compared the **time for 1 cycle** of 20 kHz audio.
- It is **OK** to **split** analogue audio.
 - Sometimes called **Parallel**
- **Long distance** (telecoms) audio used **600 Ω** characteristic impedance cable and correct terminations at each end. Such signals **could not** be paralleled.











The End